Code No: 153AB **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, September - 2021
ANALOG AND DIGITAL ELECTRONICS
(Common to CSE, IT, ITE)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the characteristics and applications of a photodiode.
b) How does the reverse saturation current of a diode varies with temperature? Explain. [7+8]

2.a) Draw a circuit diagram of series inductor filter with half wave rectifier. Explain with input and output waveforms.
b) Explain negative peak clipper with and without reference voltage. [8+7]

3.a) Explain self bias. Derive the expression for S? Why it is widely used.
b) Prove that the transistor acts as an amplifier with suitable circuit diagram. [7+8]

4. Derive the expression for A_{VS}, A_{IS}, R_I, R_o of transistor amplifier using CB configuration. [15]

5.a) Explain the operation and characteristics of N- channel JFET.
b) Discuss any two applications of FET. [8+7]

6.a) Explain about Transistor–Transistor logic. Also mention the types of output configuration.
b) Prove that AND-OR network is equivalent to NAND-NAND network. [8+7]

7.a) Give the simplest logic circuit for following logic equation where d represents don't care conditions.
$F(A,B,C,D) = \sum m(7) + d(10, 11, 12, 13, 14, 15)$.
b) Design a 32:1 multiplexer using 16:1 Mux and 2:1 multiplexer? [9+6]

8.a) Explain 4 bit parallel in serial out shift register.
b) Explain about RAM with neat sketches. [8+7]

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Used Papers SEP 2021